Notes: Kashyap, Kovrijnykh, Li, & Pavlova (AER, 2023)
Is There Too Much Benchmarking in Asset Management?

Contents

1 Big picture

- **Question.** Why is portfolio-manager compensation so often benchmarked, and is the privately chosen amount of benchmarking socially excessive?
- **Setting.** The paper embeds an optimal-contracting problem for delegated asset management inside a general-equilibrium asset-pricing model.
- **Agents.** Direct investors trade the risky asset themselves. Fund investors cannot trade the risky asset directly and hire portfolio managers. Portfolio managers choose unobservable portfolios and incur private costs of managing them.
- **Contracting friction.** The manager's portfolio is not contractible. The fund investor cannot simply tell the manager how much risky asset to hold.
- **Managerial technology.** Managers can generate expected abnormal return, or "alpha," but doing so is privately costly and noisy.
- **Privately optimal contract.** Linear compensation includes a fixed component, an absolute-performance component, and a relative-performance component. Benchmarking emerges endogenously because it partially shields the manager from market risk while preserving incentives to take costly alpha-producing positions.
- **Key general-equilibrium force.** Incentive contracts increase managers' aggregate demand for the risky asset. This pushes up the risky asset price and lowers its expected return.
- **Externality.** Each fund investor takes the asset price as given and ignores that his contract contributes to crowded trades that make other managers' incentive contracts less effective.
- **Planner result.** A constrained social planner, facing the same unobservability and linear-contract restrictions, chooses less "skin in the game" and less benchmarking than decentralized fund investors.
- **Crowding result.** Under the social contract, the risky asset price is lower, fund managers hold less of the risky asset, and asset-management costs are lower.
- **Economic takeaway.** Benchmarking is privately useful but socially excessive. The problem is not benchmarking per se; it is that benchmarking crowds trades and weakens incentive provision in equilibrium.

2 Model primitives and measurement

2.1 Investment opportunities

- There are two dates, $t = 0, 1$.
- The economy has one risky asset and one risk-free bond.
- The risky asset has payoff $\widetilde{D} \sim N(\mu, \sigma^2)$ at date 1.
- The risk-free bond pays a zero interest rate.
- The risky asset has net supply $\bar{x} > 0$.
- The date-0 price of the risky asset is p.

Interpretation: The risky asset can be interpreted as the stock market. The one-risky-asset model is chosen to make the contract-price externality transparent.

2.2 Agents

- Direct investors manage their own portfolios.
- Fund investors can only buy the bond themselves and must hire managers to trade the risky asset.

- Each manager works for one fund investor and cannot trade the risky asset personally to undo the compensation contract.
- The population shares are λ_D direct investors and λ_M managers, with $\lambda_D + 2\lambda_M = 1$ because each manager is paired with one fund investor.
- Agents have CARA utility:
$$U(W) = -e^{-\gamma W}, \qquad \gamma > 0.$$

Interpretation: CARA-normal assumptions convert the problems into mean-variance form and allow closed-form optimal contracts.

2.3 Managerial alpha technology

The manager's portfolio return is
$$r_x = x(\Delta + \widetilde{D} - p) + \varepsilon, \tag{1}$$
where $\Delta \geq 0$ is expected abnormal return and $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ is noisy alpha production. The manager pays a private management cost $x\psi$, where $\psi > 0$.

Interpretation: The key agency problem is that the manager privately bears $x\psi$, but the fund investor benefits from the alpha-producing portfolio. If $\psi = 0$ or if x were contractible, the central problem would disappear.

2.4 Linear compensation contract

The contract is
$$w = \widehat{a} r_x + b(r_x - r_b) + c = a r_x - b r_b + c, \tag{2}$$
where $r_b = \widetilde{D} - p$ is the benchmark return. The contract has three pieces:

- c: fixed compensation;
- $\widehat{a}$: sensitivity to absolute performance;
- b: sensitivity to relative performance;
- $a = \widehat{a} + b$: total "skin in the game."

Interpretation: Benchmarking means the manager is paid for outperforming the benchmark. In this model, this reduces the manager's exposure to aggregate risky-asset fluctuations while still encouraging alpha-producing positions.

2.5 What is measured in the model

- a measures incentive intensity or skin in the game.
- b measures relative-performance benchmarking.
- b/a is central because the manager's effective benchmark hedge enters portfolio choice through this ratio.
- p measures the equilibrium risky-asset price.
- x^M measures each manager's risky-asset holdings.
- ψx^M measures the cost of asset management.

Interpretation: The welfare comparison is not about whether managers can generate alpha. It is about whether privately chosen contracts generate too much aggregate demand, price pressure, and management cost.

3 Models and equilibrium specifications

3.1 Direct investor problem

A direct investor chooses x to maximize the mean-variance objective

$$\max_x \; x(\mu - p) - \frac{\gamma}{2} x^2 \sigma^2. \tag{3}$$

The first-order condition gives

$$x^D = \frac{\mu - p}{\gamma \sigma^2}. \tag{4}$$

Interpretation: Direct investors hold the standard mean-variance portfolio. They demand more of the risky asset when its expected return $\mu - p$ is high and less when volatility or risk aversion is high.

3.2 Manager problem

Given contract (a, b, c), the manager's mean-variance problem is

$$\max_x \; ax(\Delta - \psi/a + \mu - p) - b(\mu - p) + c - \frac{\gamma}{2}\left[(ax - b)^2 \sigma^2 + a^2 \sigma_\varepsilon^2\right]. \tag{5}$$

The manager's risky-asset demand is

$$x^M = \frac{\Delta - \psi/a + \mu - p}{a\gamma\sigma^2} + \frac{b}{a} = \frac{x^D}{a} + \frac{\Delta - \psi/a}{a\gamma\sigma^2} + \frac{b}{a}. \tag{6}$$

Interpretation: The manager's demand differs from a direct investor's in three ways. First, performance pay scales demand by $1/a$. Second, the manager perceives the alpha opportunity net of private cost ψ/a. Third, benchmarking creates a benchmark-hedging term b/a.

3.3 Market clearing and price

Market clearing is

$$\lambda_M x^M + \lambda_D x^D = \bar{x}. \tag{7}$$

Solving gives

$$p = \mu - \gamma\sigma^2 \Lambda \left(\bar{x} - \lambda_M \frac{b}{a}\right) + \Lambda \frac{\lambda_M}{a}\left(\Delta - \frac{\psi}{a}\right), \qquad \Lambda = \left(\frac{\lambda_M}{a} + \lambda_D\right)^{-1}. \tag{8}$$

Interpretation: Benchmarking raises the price through the term $\lambda_M b/a$. Alpha incentives also raise price when managers' alpha opportunity net of cost is attractive. This price effect is the source of crowded trades.

3.4 Fund investor problem

Define

$$y = ax - b, \qquad z = x - y. \tag{9}$$

Here y is the manager's effective risky-asset exposure and z is the fund investor's effective risky-asset exposure. The fund investor solves

$$\max_{a, b/a, c} U^F agFI \tag{10}$$

subject to the manager participation constraint

$$U^M \geq u_0, \tag{11}$$

and the manager incentive constraint

$$y = \frac{\Delta - \psi/a + \mu - p}{\gamma\sigma^2}. \tag{12}$$

Interpretation: The fund investor chooses contract terms, not the portfolio directly. The incentive constraint is the reduced-form expression of noncontractible portfolio choice.

3.5 Private first-order condition for benchmarking

The private FOC with respect to b/a is

$$\Delta - \psi + \mu - p - \gamma\sigma^2 z = 0. \tag{13}$$

Equivalently,

$$\gamma\sigma^2 b = (2a-1)(\Delta - \psi + \mu - p) + (1-a)\left(\frac{1}{a} - 1\right)\psi. \tag{14}$$

Interpretation: The optimal benchmark balances the manager's total expected surplus against variance costs. Benchmarking makes the manager invest more while helping share aggregate dividend risk.

3.6 Private optimal contract

In the privately optimal equilibrium, a^* and b^* solve

$$0 = (1-a^*)\frac{\psi^2}{\gamma\sigma^2 a^{*3}} - (2a^*-1)\gamma\sigma_\varepsilon^2, \tag{15}$$

$$b^* = (2a^*-1)\left[\bar{x} + \frac{\lambda_D}{\gamma\sigma^2}(\Delta - \psi)\right] + (1-a^*)\left[\frac{1}{a^*} - \left(\frac{\lambda_M}{a^*} + \lambda_D\right)\right]\frac{\psi}{\gamma\sigma^2}. \tag{16}$$

The equilibrium price is

$$p^* = \mu - \gamma\sigma^2\bar{x} + \lambda_M\left(2\Delta - \psi - \frac{\psi}{a^*}\right), \tag{17}$$

and each fund's risky-asset holdings are

$$x^{M*} = 2\bar{x} + \frac{\lambda_D}{\gamma\sigma^2}\left(2\Delta - \psi - \frac{\psi}{a^*}\right). \tag{18}$$

Interpretation: The recursive structure is important. First solve for skin in the game a^*, then b^*, then price and holdings. The privately optimal contract uses benchmarking because it lowers the cost of incentivizing risky alpha production.

3.7 Proposition 1: benchmarking is privately optimal

The paper proves:

$$a^* \in (1/2, 1). \tag{19}$$

If

$$\bar{x} + \frac{\lambda_D(\Delta - \psi)}{\gamma\sigma^2} > 0, \tag{20}$$

then

$$b^* > 0. \tag{21}$$

Interpretation: Perfect risk sharing would set $a = 1/2$, while selling the project to the manager would set $a = 1$. The private optimum lies between them. Benchmarking is optimal because it is a useful signal in the sense of Holmström: it filters market risk from alpha incentives.

3.8 Social planner problem

The constrained planner maximizes a weighted average of fund-investor and direct-investor utilities:

$$\max_{a,b/a,c} \ \omega_F U^F + \omega_D U^D, \tag{22}$$

subject to the same participation, incentive, and market-clearing constraints. The key difference is that the planner recognizes

$$p = \widehat{p}(a,b), \tag{23}$$

whereas each fund investor treats p as fixed.

Interpretation: The planner faces the same contracting technology as private agents. The welfare result is therefore not driven by a superior planner technology, but by internalizing a general-equilibrium pecuniary externality.

3.9 Social first-order condition for benchmarking

The planner's FOC with respect to b/a can be written as a private FOC plus an extra term involving $\partial p/\partial(b/a)$. The simplified condition is

$$\left(\Delta + \mu - p - \gamma\sigma^2 z\right)\left[1 - \frac{(1-a)\lambda_M/a}{\lambda_M/a + \lambda_D}\right] - \psi = 0. \tag{24}$$

Equivalently,

$$\Delta - \frac{\lambda_M/a + \lambda_D}{\lambda_M + \lambda_D}\psi + \mu - p - \gamma\sigma^2 z = 0. \tag{25}$$

Interpretation: From the planner's perspective, the marginal benefit of benchmarking is smaller because extra benchmark demand pushes up the asset price and lowers expected return. The perceived marginal cost is also higher because the crowding term dilutes incentives.

3.10 Socially optimal contract

In the socially optimal equilibrium, a^{**} and b^{**} solve

$$0 = (1-a^{**})\frac{\psi^2}{\gamma\sigma^2 a^{**3}}\frac{\lambda_D}{\lambda_M + \lambda_D} - (2a^{**}-1)\gamma\sigma_\varepsilon^2, \tag{26}$$

with a corresponding expression for b^{**}. The socially optimal risky-asset price is

$$p^{**} = \mu - \gamma\sigma^2\bar{x} + \lambda_M\left(2\Delta - \frac{\lambda_M/a^{**} + \lambda_D}{\lambda_M + \lambda_D}\psi - \frac{\psi}{a^{**}}\right). \tag{27}$$

Interpretation: The social contract uses the same objects as the private contract, but corrects the perceived benefit and cost of incentives to account for how contracts move prices.

3.11 Proposition 2: less benchmarking and less skin in the game

The paper's central comparison is

$$a^{**} < a^*, \tag{28}$$

and, under the same positivity condition,

$$b^{**} < b^*. \tag{29}$$

Interpretation: Fund investors privately overincentivize managers. The planner chooses less performance sensitivity and less relative-performance benchmarking because the decentralized contracts impose a negative contracting externality on other fund investors.

3.12 Proposition 3: crowded trades and excessive costs

Compared with the decentralized equilibrium, the social optimum has

$$p^{**} < p^{*}, agCrowding1 \tag{30}$$

$$x^{M**} < x^{M*}, \qquad \psi x^{M**} < \psi x^{M*}.agCrowding2 \tag{31}$$

Interpretation: Private contracts push the risky asset price too high, induce excessive manager holdings, and generate excessive asset-management costs.

4 Identification and economic design

4.1 What is identified in the model

- The paper identifies a mechanism, not an empirical treatment effect.
- The mechanism is that benchmarked incentive contracts change aggregate risky-asset demand.
- Price pressure feeds back into all fund investors' incentive problems.
- Individual contracting decisions are privately optimal but constrained inefficient.

Interpretation: The model provides a welfare benchmark for asset-management contracting: privately optimal benchmarking can be excessive even though it is individually rational.

4.2 Why benchmarking appears in contracts

- Benchmarking is useful because the benchmark return is correlated with the fund return.
- Relative performance evaluation lets the fund investor filter out market-wide risk.
- This makes it cheaper to incentivize costly alpha-producing activity.
- The model therefore rationalizes why real-world mutual fund contracts frequently include benchmark-based pay.

Interpretation: The paper does not assume benchmarking is a mistake. It first explains why it is privately optimal, then shows why the decentralized equilibrium contains too much of it.

4.3 Why the externality matters

- In complete markets, pecuniary externalities often only redistribute wealth.
- Here markets are incomplete because managerial portfolios are unobservable and contracts are restricted.
- Asset prices enter the manager's incentive constraint.
- When one fund investor benchmarks more, the resulting price pressure weakens the incentive returns available to other managers.

Interpretation: The inefficiency is not merely that the risky asset price changes. The inefficiency is that the price change relaxes or tightens incentive trade-offs in a way each individual investor ignores.

4.4 Why the planner chooses less incentive provision

- A private fund investor views the asset price as fixed.
- The planner knows that raising b/a collectively increases demand for the risky asset.

- Higher demand raises p and lowers $\mu - p$.
- Lower $\mu - p$ makes alpha-producing positions less attractive for all managers.
- Therefore, the planner chooses lower a and lower b.

Interpretation: The planner's less aggressive contract is a response to crowded trades, not to paternalism about managers' risk taking.

4.5 What the paper cannot fully prove

- The analysis is theoretical and highly stylized.
- The main model has one risky asset, one benchmark, CARA-normal preferences, and linear contracts.
- The model abstracts from endogenous entry into fund management in the baseline.
- Passive management, ESG benchmark design, and multi-asset benchmark construction are discussed as extensions or future work rather than fully developed in the baseline paper.
- The model does not estimate how large the social cost of benchmarking is in the data.

Interpretation: The paper's central value is conceptual: it shows a clean channel through which standard compensation practices can generate crowded trades and constrained inefficiency.

5 Tables, figures, and original excerpts

The paper has no empirical tables or plotted figures in the main text. The visual material below therefore consists of original equation and proposition excerpts directly cropped from the paper, grouped to save space and improve reading speed.

5.1 Original excerpt group 1: manager return and compensation contract

the interpretation of these benefits and costs in online Appendix C.

We model the costs and benefits as follows. Throughout, we will work with per-share rather than per-dollar returns. The return for a direct investor's portfolio x is given by $x(\tilde{D} - p)$. The fund manager's return is

$$r_x = x(\Delta + \tilde{D} - p) + \varepsilon, \tag{1}$$

where $\Delta \geq 0$ is the (exogenous) expected abnormal return and $\varepsilon \sim N(0, \sigma_\varepsilon)$ is a noise term. We will refer to the excess return of $x\Delta + \varepsilon$ as "alpha." The manager incurs a private portfolio-management cost $x\psi$, where $\psi > 0$ is the exogenous cost per share.

There are several key ingredients that are crucial for our results. It is essential for our mechanism that the manager's portfolio is not contractible (or unobservable), and the manager incurs a private cost of managing it (meaning that this cost is borne

(a) Manager return and private cost technology.

To provide incentives for the managers to invest in the risky asset and to generate alpha, the fund investors design compensation contracts. The managers receive compensation w from fund investors. We assume that this compensation has three parts: the first is a linear payout based on absolute performance of the manager's portfolio x, a second part that depends on the performance relative to a benchmark portfolio, and a third that is independent of performance.[13] The benchmark portfolio is one share of the risky asset. That is, the manager's compensation is given by

$$w = \hat{a}r_x + b(r_x - r_b) + c = ar_x - br_b + c, \tag{2}$$

where r_x is the performance of the manager's portfolio defined in (1) and $r_b = \tilde{D} - p$ is the performance of the benchmark portfolio.[14] The contract for a manager depends on three numbers $(\hat{a}, b, c)$—or, equivalently, (a, b, c). We refer to $\hat{a}$ as the sensitivity to absolute performance and b as the sensitivity to relative performance. Our main analysis and the intuitions that follow will be in terms of a rather than $\hat{a}$. We refer to the variable a as the manager's "skin in the game." The contract for a

(b) Linear compensation with relative performance.

Read:

- Equation (1) defines the manager's return as risky-asset exposure times market return plus alpha and noise.
- Equation (2) defines the compensation contract as absolute performance plus relative performance plus a fixed component.
- The contract converts benchmarking into a hedge against benchmark risk and a channel for incentive provision.

Economic message: Benchmarking is endogenous: it is part of the optimal contract used to motivate unobservable, costly alpha-producing portfolio choice.

5.2 Original excerpt group 2: demand functions and fund investor constraints

$$\max_x ax(\Delta - \psi/a + \mu - p) - b(\mu - p) + c - \frac{\gamma}{2}\left[(ax - b)^2\sigma^2 + a^2\sigma_\varepsilon^2\right].$$

Note that the manager receives a fraction a of the per-share abnormal return on the assets, Δ, but pays the entire cost ψ per share. (We later show that $a < 1$.)

Both the direct investors and managers take the stock price as given. Lemma 1 reports the optimal portfolio choices of the direct investors and managers arising from their optimizations, and the market-clearing asset price (for a given contract) arising from the market-clearing condition $\lambda_M x^M + \lambda_D x^D = \bar{x}$.[19]

LEMMA 1 (Portfolio Choices and Market-Clearing Price): *For a given a contract* (a, b, c),

(*i*) *the direct investors' and managers' optimal portfolio choices are as follows*:

$$x^D = \frac{\mu - p}{\gamma\sigma^2}, \tag{3}$$

$$x^M = \frac{\Delta - \psi/a + \mu - p}{a\gamma\sigma^2} + \frac{b}{a} = \frac{x^D}{a} + \frac{\Delta - \psi/a}{a\gamma\sigma^2} + \frac{b}{a}; \tag{4}$$

(*ii*) *the market-clearing price of the risky asset is*

$$p = \mu - \gamma\sigma^2\Lambda\left(\bar{x} - \lambda_M\frac{b}{a}\right) + \Lambda\frac{\lambda_M}{a}\left(\Delta - \frac{\psi}{a}\right), \tag{5}$$

where $\Lambda \equiv (\lambda_M/a + \lambda_D)^{-1}$ *modifies the market's effective risk aversion.*

[19] We define the equilibrium at the end of Section IIIB after we introduce the fund investor's problem.

(a) Lemma 1: direct demand, manager demand, and market-clearing price.

where x and z are given by

$$x = \frac{y}{a} + \frac{b}{a},$$

$$z = \left(\frac{1}{a} - 1\right)y + \frac{b}{a} = \left(\frac{1}{a} - 1\right)\frac{\Delta - \psi/a + \mu - p}{\gamma\sigma^2} + \frac{b}{a}. \tag{6}$$

Then the fund investor's problem can then be written as follows:[22]

$$\max_{a, b/a, c} U^F$$

s.t.

$$U^M \geq u_0, \tag{7}$$

$$y = \frac{\Delta - \psi/a + \mu - p}{\gamma\sigma^2}. \tag{8}$$

Constraint (7) is the manager's participation constraint, where u_0 is (the mean-variance equivalent of) the value of manager's outside option.[23] Equation (8) is the manager's (modified) incentive constraint.

An equilibrium with privately optimal contracts consists of the contract, risky asset holdings by direct investors and fund managers, and the stock price such that the agents solve their corresponding problems and the stock market clears. Appendix A contains the formal definition. We characterize this equilibrium in the next subsection.

(b) Fund investor problem, participation, and incentive constraints.

Read:

- Lemma 1 shows that the manager demand has a standard mean-variance term, an alpha-cost term, and a benchmark-hedging term.
- The market-clearing price depends directly on b/a.
- The fund investor chooses contract terms subject to manager participation and incentive compatibility.

Economic message: The contract affects the asset price because it affects manager demand. This is the bridge from optimal contracting to general equilibrium asset pricing.

5.3 Original excerpt group 3: private contract FOCs and private optimal contract

the fund. But since the manager bears the cost privately and only receives fraction a of the return, for her the effective cost is higher, which is why ψ/a appears in (8). The difference between the actual cost (ψ) and the cost perceived by the manager (ψ/a) will play an important role in the trade-off between risk sharing and incentive provision.

First, consider the fund investor's optimal choice of relative performance in the contract, b. Notice that b enters the fund investor's and manager's problems only though b/a. The FOC with respect to b/a is given by[27]

$$\frac{\partial\left(U^F + U^M\right)}{\partial(b/a)} = \Delta - \psi + \mu - p - \gamma\sigma^2 z = 0. \tag{9}$$

This condition captures the fact that an increase in b/a makes the manager invest more in the risky stock. Therefore, the optimal level of b will be the one that balances a marginal increase in the total expected surplus, $\Delta - \psi + \mu - p$, with the marginal increase in the variance, $\sigma^2 z$.

Substituting out z using (6), equation (9) can be rewritten as[28]

$$\gamma\sigma^2 b = (2a - 1)(\Delta - \psi + \mu - p) + (1 - a)\left(\frac{1}{a} - 1\right)\psi. \tag{10}$$

(a) Private FOC for benchmarking.

tion, but is also costly because it exposes the manager to too much risk.

The following lemma summarizes the closed-form expressions for the equilibrium contract, as well as the expressions for the equilibrium price and stock holdings:

LEMMA 2: *In the equilibrium with privately optimal contracts,*

(*i*) a^* *and* b^* *solve*

$$0 = (1 - a^*)\frac{\psi^2}{\gamma\sigma^2 a^{*3}} - (2a^* - 1)\gamma\sigma_\varepsilon^2, \tag{12}$$

$$b^* = (2a^* - 1)\left[\bar{x} + \frac{\lambda_D}{\gamma\sigma^2}(\Delta - \psi)\right] + (1 - a^*)\left[\frac{1}{a^*} - \left(\frac{\lambda_M}{a^*} + \lambda_D\right)\right]\frac{\psi}{\gamma\sigma^2}; \tag{13}$$

(*ii*) *the risky asset's price is*

$$p^* = \mu - \gamma\sigma^2\bar{x} + \lambda_M\left(2\Delta - \psi - \frac{\psi}{a^*}\right); \tag{14}$$

and each fund's risky asset holdings are

$$x^{M*} = 2\bar{x} + \frac{\lambda_D}{\gamma\sigma^2}\left(2\Delta - \psi - \frac{\psi}{a^*}\right). \tag{15}$$

(b) Lemma 2: privately optimal contract, price, and holdings.

Read:

- The private FOC makes benchmarking balance expected surplus against variance costs.

- Lemma 2 pins down the private equilibrium recursively: first a^*, then b^*, then p^* and x^{M*}.
- The price contains the manager-driven term $\lambda_M(2\Delta - \psi - \psi/a^*)$.

Economic message: Private benchmarking increases managers' demand and raises the stock price, but private fund investors ignore the equilibrium feedback on everyone else's contracts.

5.4 Original excerpt group 4: benchmarking is optimal and the planner's FOC

that the equilibrium level a^* is strictly between $1/2$ (perfect risk sharing) and 1 (private and social costs coincide). Also note that as σ_ε^2 goes to zero, a^* approaches 1, and the allocation approaches the first-best one (see Lemma 5 in Appendix A.) Indeed, it is crucial for our results that the fund investor does not "sell the project" to the manager, i.e., $a^* < 1$. As an alternative to the assumption of $\sigma_\varepsilon^2 > 0$, there are other modeling choices that would ensure that $a^* < 1$, for example, a lower-bound on c, the constant part of the contract.

As long as $\bar{x} + \lambda_D(\Delta - \psi)/(\gamma\sigma^2) > 0$, all the terms on the right-hand side of (13) are positive. This condition is satisfied either if $\Delta - \psi > 0$ (the expected abnormal return exceeds the cost of managing the portfolio), or if the net supply of the stock $\bar{x}$ is large enough. This brings us to our first main result.

PROPOSITION 1 (Benchmarking Is Optimal): *Consider the equilibrium with privately optimal contracts.*

(*i*) *The equilibrium value of the "skin in the game" satisfies* $a^* \in (1/2, 1)$.

(*ii*) *Suppose that* $\bar{x} + \lambda_D(\Delta - \psi)/(\gamma\sigma^2) > 0$. *Then benchmarking is optimal, that is,* $b^* > 0$.

Part (ii) of Proposition 1 is essentially a version of Holmström's (1979) famous sufficient-statistic result—the use of an additional signal (in this case, the benchmark return) helps the contract designer provide incentives to the manager in a more effective way. While Holmström's result suggests that b^* is different from zero in

(a) Proposition 1: private benchmarking is optimal.

As before, this problem can be equivalently rewritten in terms of mean-variance preferences.[32] Define $U^D = x_{-1}^D p + x^D(\mu - p) - \gamma x^{D2}\sigma^2/2$. Then the social planner's problem is

$$\max_{a,\, b/a,\, c} \omega_F U^F + \omega_D U^D,$$

subject to (3), (5), (7), and (8).

The social planner's FOC with respect to b/a is

$$0 = \underbrace{\left[\omega_F\left(x_{-1}^F - x^M\right) + \omega_D\left(x_{-1}^D - x^D\right)\right]\frac{\partial p}{\partial(b/a)}}_{\text{distributive pecuniary externality}} \tag{16}$$

$$+\ \omega_F\left[\underbrace{\frac{\partial\left(U^F + U^M\right)}{\partial(b/a)}}_{\text{private FOC}} + \underbrace{\frac{\partial U^F}{\partial y}\frac{\partial y}{\partial p}\frac{\partial p}{\partial(b/a)}}_{\text{contracting pecuniary externality}}\right].$$

The terms in the first line of (16) capture what Davila and Korinek (2018) call "distributive effects" or "distributive pecuniary externality." Depending on the initial endowments and the Pareto weights, the social planner has incentives to use benchmarking to move the price so as to benefit one or the other party based on this distributive motive. We discuss the distributive effects in Remark 1 at the end of the next section. Our focus is on the "contracting pecuniary externality" that acts through the price entering the manager's incentive constraint. To isolate the planner's motive to correct the contracting externality from the distributive motive, we want to neutralize the latter. To do this, we set the Pareto weights equal to the population weights, $\omega_F = \lambda_M$ and $\omega_D = \lambda_D$.[33] Then by market clearing, $\omega_F\left(x_{-1}^F - x^M\right) + \omega_D\left(x_{-1}^D - x^D\right) = 0$, so the term in the first line of (16) is zero. (See Davila and Korinek 2018 for further discussion.)

Rewriting the term in the second line of (16) yields

$$0 = \underbrace{\left(\Delta - \psi + \mu - p - \gamma\sigma^2 z\right)\underbrace{\frac{\partial y}{\partial(b/a)}}_{=1}}_{\text{private FOC}} \tag{17}$$

(b) Planner FOC and contracting externality.

Read:

- Proposition 1 says $a^* \in (1/2, 1)$ and $b^* > 0$ under a natural condition.
- The planner's FOC explicitly contains a pecuniary externality term.
- The planner recognizes that changing b/a changes the equilibrium price.

Economic message: Benchmarking is privately rational. The social problem arises only after aggregating many privately rational benchmarked contracts.

5.5 Original excerpt group 5: social optimal contract and the central welfare result

ignore.

Consider the term $(\partial y/\partial p)\left[\partial p/\partial(b/a)\right]$. The term $\partial y/\partial p = -1/(\gamma\sigma^2)$ captures the fact that the manager's demand function is downward sloping. The term $\partial p/\partial(b/a) = \gamma\sigma^2\Lambda\lambda_M$ reflects the fact that the higher the value b/a collectively used by all fund investors, the more crowded the trades, and the higher the stock price. The product of the two, $(\partial y/\partial p)\left[\partial p/\partial(b/a)\right] = -\Lambda\lambda_M = -\lambda_M/(\lambda_M/a + \lambda_D)$ captures the fact that the general equilibrium effect of contracts on the risky asset's price reduces the effectiveness of b/a in incentivizing the manager to hold more of the risky asset. Hence (17) becomes

$$(\Delta + \mu - p - \gamma\sigma^2 z)\left[1 - \frac{(1-a)\lambda_M/a}{\lambda_M/a + \lambda_D}\right] - \psi = 0 \tag{18}$$

or

$$\Delta - \underbrace{\frac{\lambda_M/a + \lambda_D}{\lambda_M + \lambda_D}\psi}_{\text{cost from the planner's perspective}} + \mu - p - \gamma\sigma^2 z = 0. \tag{19}$$

Similar to the fund investors, the planner trades off the benefits and costs of inducing the agent to invest in the risky asset. Fund investors think of the benefit as the usual mean-variance consideration given by $(\Delta + \mu - p - \gamma\sigma^2 z)$, and the cost as ψ. For the planner, the benefit is smaller than for the fund investors, because she realizes that benchmarking inflates the risky asset's price and thus reduces its expected return. Put differently, due to this crowded-trades effect, the cost is higher for the same units of benefit: the cost is $(\lambda_M/a + \lambda_D)/(\lambda_M + \lambda_D)\psi$ in (19) versus ψ in (9). (This difference in the perceived costs will show up in our further comparisons between the socially and privately optimal contracts.) So, from the planner's point of view, incentive provision is less beneficial/more expensive, which, as we will see, will make her do less of it.

Substituting for z, we obtain the planner's counterpart to equation (10):

$$\gamma\sigma^2 b = (2a-1)\left[\Delta - \frac{\lambda_M/a + \lambda_D}{\lambda_M + \lambda_D}\psi + \mu - p\right] + (1-a)\left[\frac{1}{a} - \frac{\lambda_M/a + \lambda_D}{\lambda_M + \lambda_D}\right]\psi. \tag{20}$$

Compared to (10), the cost ψ is again replaced with $(\lambda_M/a + \lambda_D)/(\lambda_M + \lambda_D)\psi$. Finally, substituting in the equilibrium price, (5), yields the fixed-point value of b

(a) Planner's perceived cost and FOC.

Compare this equation to its analog in the case with privately optimal contracts, (11). Notice that the benefit of incentive provision captured by the first term in (22) is smaller than the corresponding term in (11). As a result, the planner will choose a lower a than fund investors will. We will formalize this result later in Proposition 2.

The following lemma presents the resulting expressions for the equilibrium contract, price, and risky-asset holdings in closed form.[34]

LEMMA 4: *In the equilibrium with socially optimal contracts,*

(*i*) *a^{**} and b^{**} solve*[35]

$$0 = (1-a^{**})\frac{\psi^2}{\gamma\sigma^2 a^{**3}}\frac{\lambda_D}{\lambda_M + \lambda_D} - (2a^{**}-1)\gamma\sigma_\varepsilon^2, \tag{23}$$

$$b^{**} = (2a^{**}-1)\left[\bar{x} + \frac{\lambda_D}{\gamma\sigma^2}(\Delta - \psi)\right] + (1-a^{**})\left[\frac{1}{a^{**}} - \frac{\lambda_M/a^{**} + \lambda_D}{\lambda_M + \lambda_D}\right]\frac{\psi}{\gamma\sigma^2}; \tag{24}$$

(*ii*) *the risky asset's price is*

$$p^{**} = \mu - \gamma\sigma^2\bar{x} + \lambda_M\left(2\Delta - \frac{\lambda_M/a^{**} + \lambda_D}{\lambda_M + \lambda_D}\psi - \frac{\psi}{a^{**}}\right); \tag{25}$$

[34] See Appendix A for the formal definition of the equilibrium with socially optimal contracts.
[35] From (23), $1/2 \le a^{**} < 1$, where the first inequality is strict so long as $\lambda_D > 0$.

(b) Lemma 4: socially optimal contract and price.

Read:

- The planner's condition modifies the private benchmarking FOC by accounting for how price pressure lowers the expected return.
- Lemma 4 gives the social analog of Lemma 2.
- Compared with the private case, the planner treats incentive provision as more socially costly because it crowds trades.

Economic message: The social contract reduces benchmarking and incentives because equilibrium price pressure dilutes alpha incentives for all managers.

5.6 Original excerpt group 6: less benchmarking, lower prices, and lower costs

(26) $$x^{M**} = 2\bar{x} + \frac{\lambda_D}{\gamma\sigma^2}\left(2\Delta - \frac{\lambda_M/a^{**} + \lambda_D}{\lambda_M + \lambda_D}\psi - \frac{\psi}{a^{**}}\right).$$

Equations (23)−(26) are the analogs of (12)−(15) and have the same recursive structure. As expected, the two sets of equations coincide when $\lambda_M = 0$, and hence there is no externality. But so long as there are managers, the socially and privately optimal contracts are different. Proposition 2 below reveals how exactly they compare to each other.

We are now ready to present the central result of the paper.

PROPOSITION 2 (Socially versus Privately Optimal Contracts): *Compared to the privately optimal contract, the socially optimal contract involves*

(*i*) *less "skin in the game," that is,* $a^{**} < a^{*}$;

(*ii*) *less benchmarking, that is,* $b^{**} < b^{*}$, *if* $\bar{x} + \lambda_D(\Delta - \psi)/(\gamma\sigma^2) > 0$.[36]

As we have seen in our analysis, the use of contracts inflates the risky asset's price and thus *reduces the marginal benefit of incentive provision* for everyone else. The social planner internalizes this effect, and opts for less incentive provision than fund investors.

As a special case that helps make the point very clearly, suppose there are no direct investors, $\lambda_D = 0$. In this case, each fund will hold exactly $2\bar{x}$ shares and the total alpha in the economy is fixed, equal to $2\bar{x}\Delta$. The planner understands that

(a) Proposition 2: social contract uses less a and less b.

able to say that privately optimal contracts deliver excessive incentive provision.

We now show that excessive incentive provision and excessive benchmarking in private contracts give rise to crowded trades and excessive costs.

PROPOSITION 3 (Crowded Trades and Excessive Costs of Asset Management): *Compared to the equilibrium with privately optimal contracts, in the equilibrium with socially optimal contracts*

(*i*) *the risky asset's price is lower,* $p^{**} < p^{*}$;

(*ii*) *the fund's risky asset holdings are lower,* $x^{M**} < x^{M*}$, *and the managers' costs are lower,* $\psi x^{M**} < \psi x^{M*}$.

As Proposition 3 shows, excessive use of incentive contracts by fund investors inflates the risky asset's price and reduces its expected return per share. In addition, the managers overinvest in the stock market and the costs of asset management are excessively high. Our model thus contributes to the debate on whether costs of asset management are excessive and whether returns delivered by the managers justify these costs.

We close the analysis with a few final remarks about the model.

(b) Proposition 3: lower price, holdings, and costs in the social optimum.

Read:

- Proposition 2 states $a^{**} < a^{*}$ and, under the standard positivity condition, $b^{**} < b^{*}$.
- Proposition 3 states $p^{**} < p^{*}$, $x^{M**} < x^{M*}$, and $\psi x^{M**} < \psi x^{M*}$.
- These are the paper's welfare consequences: less benchmarking, lower price pressure, less risky-asset crowding, and lower management cost.

Economic message: The decentralized equilibrium has too much benchmarking because each investor ignores that his contract reduces the effectiveness of benchmarking for others.

6 Short summary

- The paper studies benchmarked compensation in delegated asset management.
- Managers can generate alpha but must privately incur costs and cannot have their portfolios directly contracted upon.
- Linear incentive contracts include absolute performance, relative performance, and a fixed payment.
- Benchmarking is privately optimal because it shields managers from benchmark risk and makes alpha incentives cheaper.
- Manager demand includes a benchmark-hedging component proportional to b/a.
- In general equilibrium, benchmarked contracts push up risky-asset demand and raise the asset price.
- Higher prices lower expected returns and reduce the effectiveness of incentive contracts for other managers.
- Individual fund investors ignore this price effect.
- A constrained planner internalizes it and chooses less skin in the game and less benchmarking.
- The socially optimal equilibrium has a lower risky-asset price, lower fund-manager risky-asset holdings, and lower asset-management costs.
- The result is a constrained inefficiency created by a pecuniary externality operating through incentive constraints.
- The model rationalizes benchmarking while simultaneously showing that decentralized asset management can produce too much of it.

7 Conclusion

7.1 Core conclusion

Kashyap, Kovrijnykh, Li, and Pavlova show that benchmarking can be both privately optimal and socially excessive. It is privately optimal because it helps fund investors design incentive contracts for managers whose portfolios are unobservable and whose alpha production is costly. It is socially excessive because benchmarked contracts crowd into the risky asset, raise its price, and weaken the incentive value of similar contracts elsewhere in the economy.

7.2 Economic intuition

The central intuition is that benchmarked pay makes managers want to hold the benchmark. This demand is individually rational, but when many managers do it, the risky asset becomes expensive. A higher price lowers the expected return that supports incentive provision. Each fund investor ignores this feedback, so the decentralized equilibrium overuses benchmarking and overpays for incentive provision.

7.3 Takeaway

The paper reframes benchmarking as a general-equilibrium contracting problem. Benchmarking is not merely a measurement device in compensation contracts; it is also a source of price pressure and crowded trades. The asset-management industry may therefore provide too much relative-performance benchmarking and too much costly active management from a social perspective.